

APPM 4/5390 Project - Spiral Waves and Defibrillation

Anil Ambrosi, Zane Perry

April 2026

1 Biological Problem

Spiral-wave dynamics in excitable media are closely connected to life-threatening cardiac arrhythmias, including tachycardia and fibrillation. In cardiac tissue, these rotating waves act as self-sustained sources of reentrant excitation, disrupting the normal coordinated propagation of electrical activity through the heart. Conventional defibrillation can terminate this activity, but it typically requires large-amplitude shocks, especially in external defibrillation, and these shocks are painful and may damage tissue [1]. Because of this, an important biological and clinical question is whether spiral-wave-mediated arrhythmias can be eliminated with smaller, more targeted interventions.

The paper we study approaches this question through the topology of excitable media. In the framework of DeTal, Kaboudian, and Fenton, the medium is divided into excited and refractory regions using level-set contours. Spiral-wave tips are identified as phase singularities, occurring at points where the relevant excitation and recovery contours intersect [1]. This point of view gives a topological description of spiral waves: rather than viewing fibrillation only as complicated irregular activity, it becomes possible to track the singularities that organize the motion of the waves.

The central idea of the paper is that these singularities can be moved and eliminated by a carefully designed stimulus applied along the refractory back. Stimulating the refractory-back contour can instantaneously relocate, or “teleport,” a spiral tip, allowing opposite-chirality singularities to annihilate or allowing a singularity to be driven to the boundary of the domain [1]. This provides a possible mechanism for low-energy defibrillation, since the stimulus is targeted to the geometry responsible for maintaining reentry rather than applied indiscriminately across the entire tissue.

Our project reproduces this mechanism in a two-dimensional FitzHugh-Nagumo reaction-diffusion model, a standard reduced model for excitable media that captures essential features of excitation, recovery, and wave propagation while remaining computationally tractable [2]. After reproducing representative spiral-wave and teleportation results, we extend the study by varying the model parameters to determine how changes in excitability, recovery, and wave structure affect spiral formation, persistence, and controllability. In this way, the project connects the topological control mechanism proposed in the reference paper to the broader biological question of what properties of excitable tissue promote or suppress reentrant spiral dynamics.

2 Model

We use a modified Reaction-Diffusion/Fitzhugh-Nagumo model:

$$\frac{\partial u}{\partial t} = -I_{\text{ion}}(u, v) + I_{\text{stim}} + D\nabla^2 u, \quad (1a)$$

$$\frac{\partial v}{\partial t} = \epsilon(\beta u - \gamma v - \delta), \quad (1b)$$

$$I_{\text{ion}}(u, v) = u(u - \alpha)(u - 1) + v. \quad (1c)$$

Here $u(x, y, t)$ is the fast excitation variable and $v(x, y, t)$ is the slower recovery variable, and I_{ion} represents the total ionic current. In the cardiac interpretation, u plays the role of a transmembrane-voltage-like

quantity, while v represents a lumped recovery process rather than a specific ion channel. In our model, the three equations are dimensionless and have images contained in $[-1, 1]$. I_{stim} represents an external stimulus term, which we will be using to simulate a heartbeat and defibrillation. D , α , β , γ , δ , and ϵ are all constant parameters representing conductivity, equilibrium ionic current, ion channel opening from voltage difference, negative ion channel feedback, ion channel decay, and a time scale on the ion channel, respectively. Also note that this model can exist in n dimensions, making extensions to higher dimensions seem natural.

A summary of parameter and variable definitions is shown in the table below:

u	Voltage difference across cell membrane
v	Ion channel opening signal
I_{ion}	Total ionic current
I_{stim}	External Stimulus
D	Conductivity (Diffusion Speed)
α	Equilibrium Ionic Current at 0 v
β	Ion channel opening stimulus from voltage difference
γ	Ion channel exponential decay
δ	Ion channel constant decay
ϵ	Ion channel rescaling

Qualitatively, this set of equations combines the conventional Fitzhugh-Nagumo equations, which model a single cardiac cell's activity through a feedback variable and a delayed voltage variable, with a diffusion term that models the spatial spread of voltage through interconnected cells on a macro scale. This results in waves of voltage, or "action potential", which propagate throughout the domain.

3 Numerical Simulations

The main numerical decision was to evolve the excitation variable u with an alternating-direction implicit (ADI) scheme, while updating the recovery variable v explicitly. This choice follows the structure of the FitzHugh-Nagumo system itself: the diffusion term appears only in the u equation, while the v equation is purely local in space. Treating v explicitly therefore avoids solving an unnecessary linear system for the recovery variable, while treating diffusion in u implicitly improves stability and allows larger time steps than a fully explicit two-dimensional method.

More specifically, each time step is split into two one-dimensional implicit diffusion solves. In the first half-step, the x -direction diffusion is treated implicitly and the y -direction diffusion explicitly; in the second half-step, the roles are reversed. For the FitzHugh-Nagumo model, Ji and Fenton show that the resulting 2D ADI method is unconditionally stable and has local truncation error

$$O(dt^2 + D dx^2 + D dy^2),$$

so it retains second-order accuracy in time and second-order accuracy in space for the diffusion discretization while remaining computationally efficient [2].

Zero-flux boundary conditions were imposed on all sides of the domain. This choice is appropriate for excitable-media simulations when one wants no diffusive current crossing the boundary, and it is consistent with the boundary treatment used in standard cardiac reaction-diffusion simulations [2]. In the discrete Laplacian, this gives modified formulas at the edges while preserving the same interior centered second-order finite-difference structure.

Because the coefficients in the implicit diffusion operators do not change in time, the corresponding tridiagonal matrices were assembled once at the beginning of the simulation and reused throughout. The code stores these matrices in banded form and solves the resulting linear systems using SciPy routines for banded linear systems [3]. The recovery variable was updated explicitly after evaluating the reaction terms at the current time level. Since the v equation contains no spatial derivatives, this update is inexpensive and keeps the numerical method aligned with the natural fast-slow structure of the model. The overall PDE step can therefore be viewed as an operator split between local reaction/recovery dynamics and an ADI diffusion solve for the voltage-like variable.

4 Computational Results

4.1 Reproduction of Results

The following parameter values were used for the initial simulations:

$$\alpha = 0.1, \quad \beta = 0.3, \quad \gamma = 1.0, \quad \epsilon = 0.010, \quad \delta = 0, \quad D = 0.0003.$$

They differ slightly from the values used in the source paper due to differences in the domain and initial values used for forced stimulation, as well as ease of presentation. All of the code used to generate the following results can be found at the following repository

<https://github.com/anilambrosi/Spiral-Wave-Modeling--APPM4390-Project->

4.1.1 Spiral Wave Formation

We begin by comparing two mechanisms for generating spiral waves in the FitzHugh–Nagumo medium. The first follows the topological framework of the main reference paper. In that construction, a traveling wave is first created, and then a localized stimulus is applied along the refractory back. Because the refractory back separates tissue that can still be excited from tissue that remains refractory, this localized excitation produces unidirectional propagation and generates a pair of phase singularities of opposite chirality. These develop into a double-spiral configuration, as shown in Figure 1 [1].

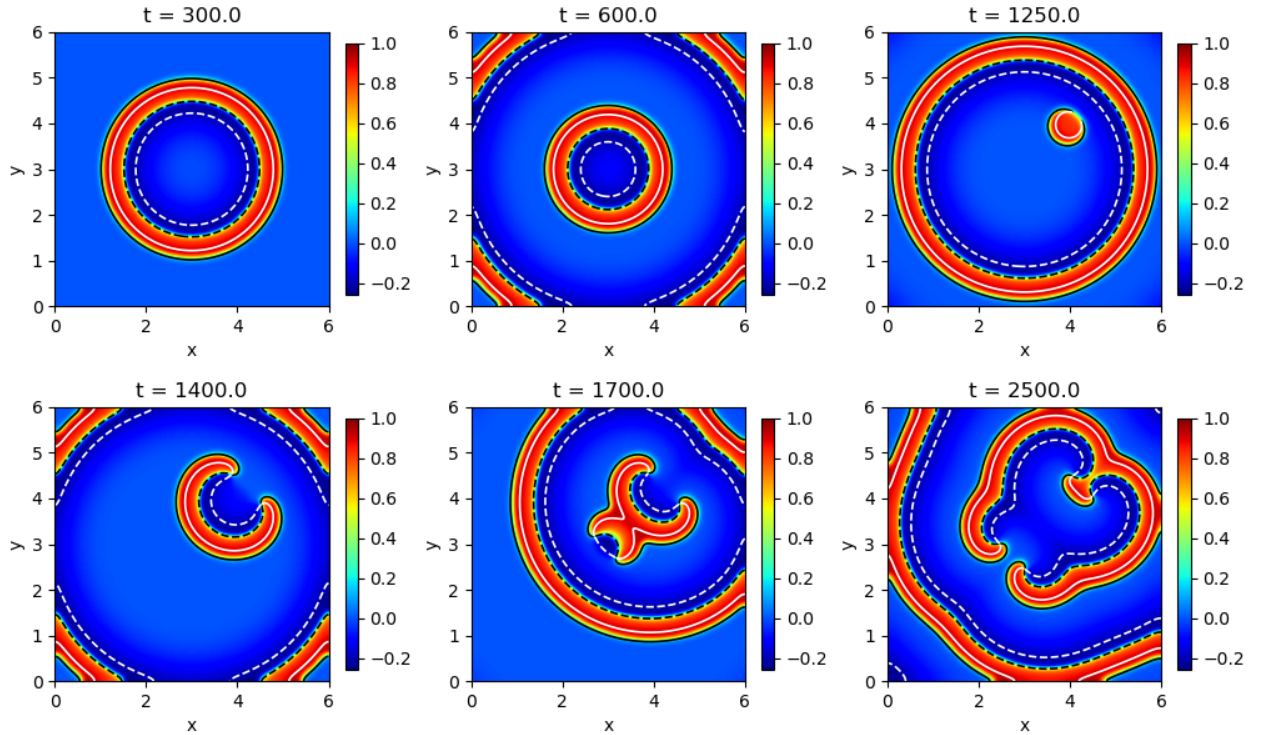


Figure 1: Spiral-wave formation by localized stimulation of the refractory back, producing a pair of spiral singularities with opposite chirality.

The second mechanism follows the numerical construction used in the reaction–diffusion simulations literature. In this approach, a plane wave is first launched across the domain, and then part of the excited wave is reset, breaking the spatial symmetry of the propagating front. The remaining broken wave curls inward and forms a single spiral wave, shown in Figure 2. Unlike the first mechanism, this construction produces a single isolated spiral and is best understood as a computational symmetry-breaking procedure rather than a direct model of physiological initiation [2].

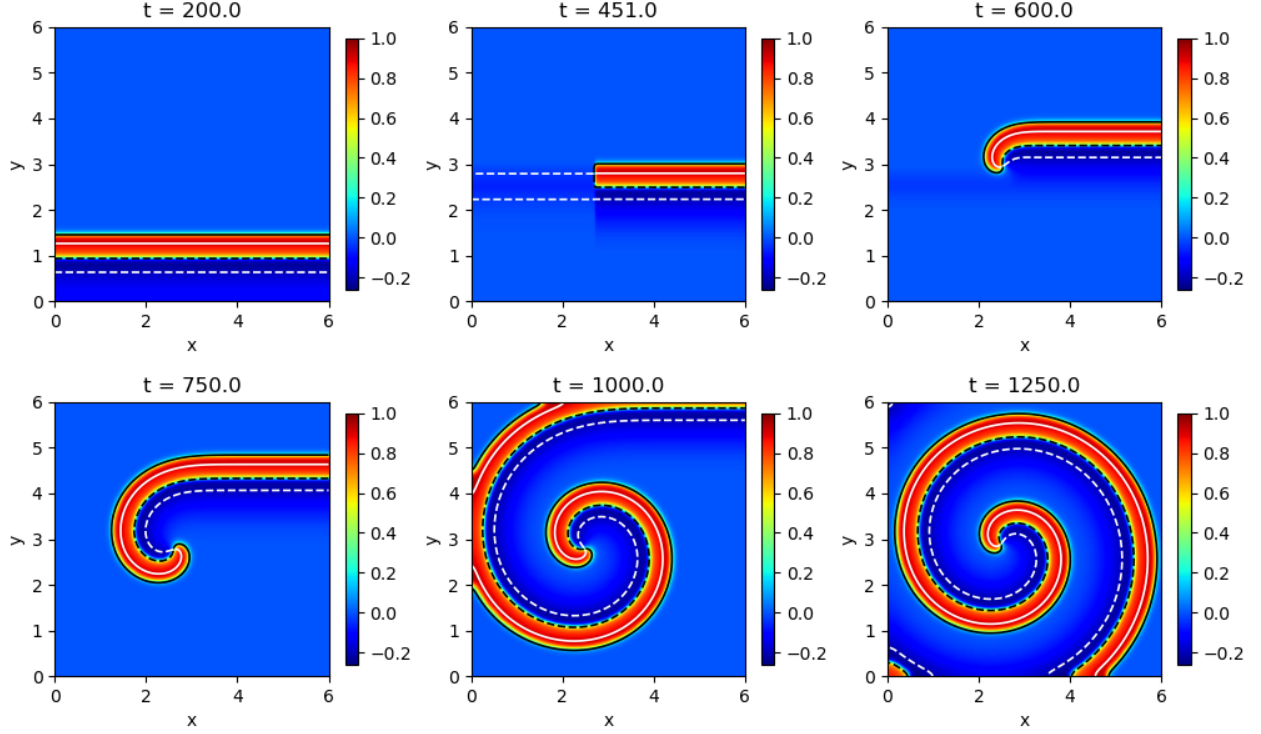


Figure 2: Spiral-wave formation by resetting part of a propagating wave, breaking symmetry and producing a single spiral of one chirality.

The refractory-back stimulation strategy more closely reflects how abnormal excitation in cardiac tissue can develop into reentrant spiral activity, since it is based on localized excitation interacting with partially recovered tissue. For that reason, it is the more biologically motivated initiation mechanism and will be returned to later in the discussion. For clarity of presentation, however, much of the analysis below focuses on the single-spiral construction, since its geometry is simpler and the resulting dynamics are easier to visualize and interpret.

4.1.2 Naive Defibrillation

As a baseline comparison, we next consider naive defibrillation by applying a uniform stimulus across the entire computational domain. In this setting, every point in the tissue is perturbed simultaneously with the same stimulus amplitude, without reference to the underlying contour geometry or the locations of the spiral tips. The purpose of this experiment is to provide a direct comparison between untargeted global stimulation and the more structured control strategies considered later.

Figures 3 and 4 show examples of whole-domain stimulation at the threshold strength required to eliminate the reentrant activity present in the medium. Below this threshold, the perturbation is insufficient to destroy all surviving wave fronts, and spiral activity persists or re-forms after the shock. At and above the threshold, the applied stimulus is strong enough to eliminate the reentrant waves and prevent their immediate reinitiation. This threshold therefore provides a natural baseline for comparing the efficiency of global and targeted stimulation strategies.

To compare shocks of different amplitudes and spatial supports, we introduce a normalized shock-burden metric

$$S = a \cdot p,$$

where $a \in [0, 1]$ is the normalized stimulus amplitude and $p \in [0, 1]$ is the fraction of the domain to which the stimulus is applied. By construction, $S \in [0, 1]$. This quantity is not intended to represent the true physical energy delivered by a clinical defibrillator. Instead, it serves as a simple computational measure of

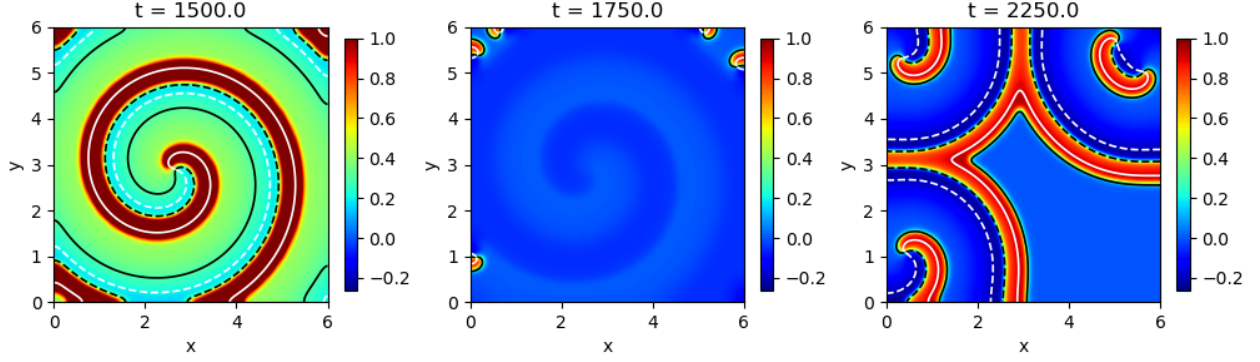


Figure 3: Whole-domain stimulation below or near the success threshold, illustrating the sensitivity of naive defibrillation to stimulus strength.

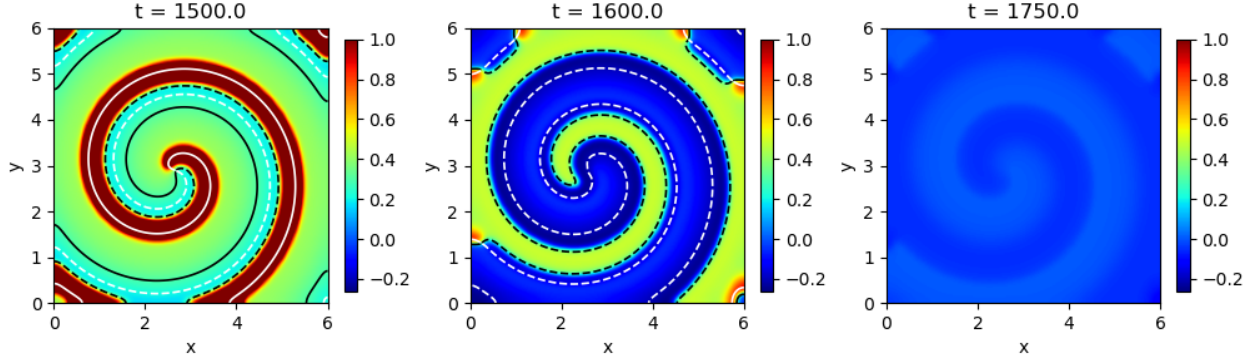


Figure 4: Whole-domain stimulation at the threshold strength required to terminate the reentrant wave pattern.

how much stimulation is being imposed on the medium, accounting simultaneously for both the strength of the perturbation and the amount of tissue affected.

For naive whole-domain defibrillation, we have $p = 1$, so the normalized shock burden is equal to the stimulus amplitude itself. For targeted shocks, however, the stimulated fraction of the domain can be much smaller, so two interventions with similar amplitude may have very different overall burdens. This makes S a useful comparative quantity when evaluating whether a designed localized stimulus can terminate spiral activity more efficiently than a uniform domain-wide perturbation.

Although whole-domain stimulation provides a useful baseline, it is not guided by the geometric structure of the reentrant waves. As a result, successful termination generally requires a sufficiently large perturbation to suppress activity everywhere at once. This motivates the later sections, where we instead use the contour structure of the excitable medium to design more spatially selective interventions.

4.1.3 Teleportation

A central mechanism in the main paper is that a spiral wave can be instantaneously relocated by applying a stimulus along its refractory back. Since the refractory back separates tissue that is able to respond to excitation from tissue that remains refractory, stimulating along this contour does not simply erase the existing spiral. Instead, it replaces the stimulated portion of the refractory back with a new excited wave segment, so that the phase singularity reforms at the far end of the applied stimulus. In this sense, the spiral tip is effectively “teleported” from its original location to the endpoint of the stimulated region [1].

This mechanism is shown in Figure 5. The applied stimulus follows the refractory-back contour of the spiral, and after stimulation the original tip no longer remains at its initial position. Rather, the singularity

reappears at the terminal end of the excited segment, producing the appearance that the spiral has moved abruptly across the domain. The key point is that this is not ordinary drift caused by subsequent evolution, but a direct topological consequence of how the contour geometry is altered by the stimulus.

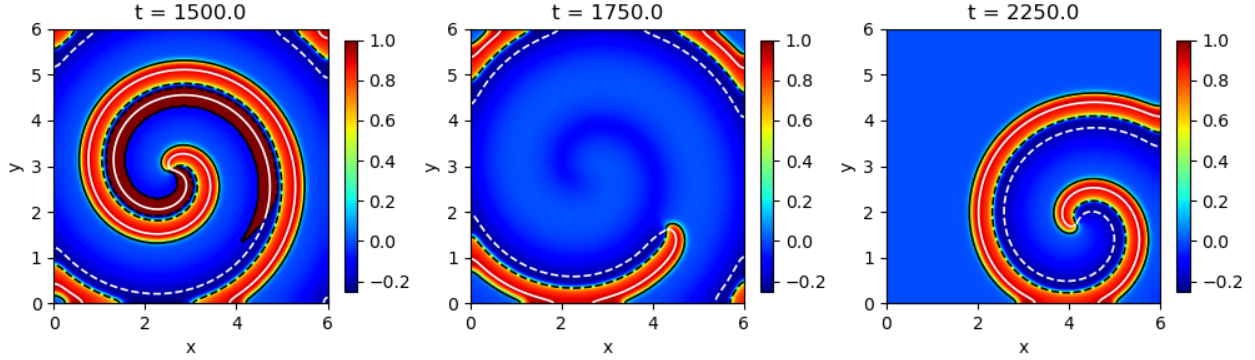


Figure 5: Teleportation of a spiral wave by stimulating along its refractory back. The singularity reforms at the end of the stimulated segment, causing the spiral tip to relocate across the domain.

This phenomenon is important because it provides the mechanism underlying the targeted control strategy. If a spiral tip can be moved in a predictable way along the refractory-back contour, then the stimulus can be designed so that the relocated singularity is brought either to the boundary or into contact with a singularity of opposite chirality. In either case, the reentrant wave can be eliminated without requiring a uniform shock to the entire domain. Teleportation therefore serves as the intermediate step between simply observing spiral waves and designing stimuli that actively annihilate them.

4.1.4 Targeted Defibrillation

The teleportation mechanism extends naturally from relocating a single spiral tip to eliminating reentrant activity altogether. Rather than applying stimulus only to a short segment of the refractory back in order to move the singularity, we instead stimulate the full contour-based region identified from the current state. In this case, the stimulus is designed so that all spiral tips are forced to relocate simultaneously along the refractory-back geometry. When this relocation brings singularities of opposite chirality together, or drives them into the boundary, the reentrant activity is destroyed [1].

Figure 6 shows this targeted defibrillation strategy. Here the stimulus was applied with normalized amplitude

$$a = 0.6,$$

and it covered a fraction

$$p = 0.20552$$

of the domain. Using the normalized shock-burden metric introduced above, this gives

$$B_{\text{shock}} = ap = 0.6 \cdot 0.20552 = 0.123312.$$

Thus, successful termination was achieved while directly stimulating only about 20.6% of the domain, with an overall shock burden substantially below that of a comparable whole-domain perturbation.

This example illustrates the main advantage of the targeted approach. A uniform domain-wide stimulus attempts to suppress activity everywhere at once, so its success depends primarily on brute-force amplitude. By contrast, the designed stimulus uses the geometry of the excitable and refractory contours to act specifically on the structures sustaining reentry. The benefit is not simply that less of the domain is shocked, but that the shocked region is chosen to alter the topological organization of the wave pattern in a way that promotes annihilation rather than survival.

In this sense, targeted defibrillation is not a separate mechanism from teleportation, but its full realization. Teleportation shows that a spiral tip can be moved by stimulating along the refractory back; targeted

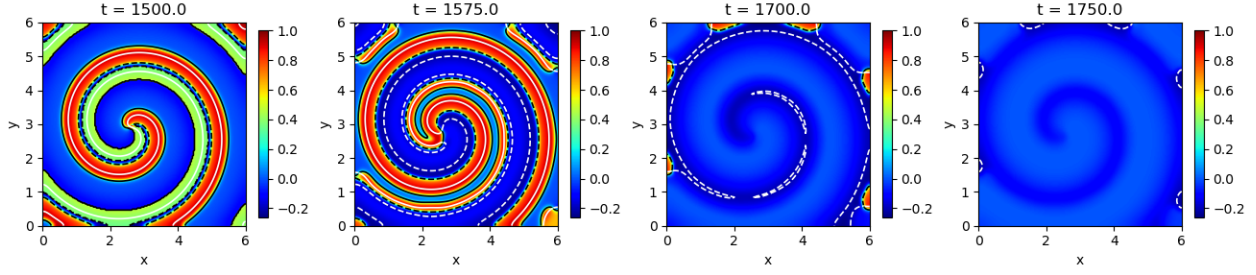


Figure 6: Targeted defibrillation by stimulation of the contour-based refractory-back region. The stimulus has amplitude 0.6 and is applied to 20.552% of the domain, giving normalized shock burden $B_{\text{shock}} = 0.123312$.

defibrillation uses that same principle on the full relevant contour set so that all surviving singularities are removed in a single designed intervention. This is precisely the distinction between an untargeted shock, which must be strong enough to overwhelm the entire medium, and a structured low-burden intervention, which succeeds by exploiting the topology of the wave pattern itself.

4.1.5 Complex Wave Patterns

To show that the contour-based strategy is not limited to a single isolated spiral, we next consider more complex reentrant states generated by the refractory-back stimulation mechanism described earlier. Unlike the symmetry-breaking construction used to create a single spiral for visualization purposes, this initiation method produces a more biologically motivated wave pattern, since localized excitation of the refractory back naturally generates pairs of opposite-chirality singularities and leads to more complicated interactions among the resulting waves. As these waves evolve, the domain develops a more intricate reentrant structure than in the single-spiral case.

Figure 7 shows a representative example of such a complex wave pattern. Compared to the single-spiral states used in the earlier sections, the geometry here is more irregular, with multiple interacting wave fronts and a less transparent overall organization. This makes the example especially useful as a test of whether the targeted defibrillation mechanism remains effective when the underlying dynamics are closer to the multispiral behavior associated with fibrillatory activity.

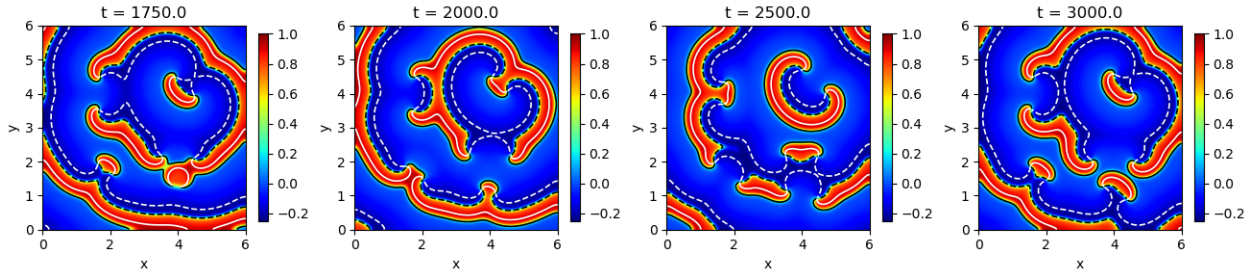


Figure 7: Complex reentrant wave pattern generated by refractory-back stimulation. This state contains a more intricate arrangement of interacting spiral activity than the isolated single-spiral examples considered earlier.

Figure 8 shows that the same targeted contour-based stimulation strategy can still eliminate this more complicated state. In this case, the designed defibrillation stimulus was applied with normalized amplitude

$$a = 0.62,$$

over a stimulated domain fraction

$$p = 0.19865.$$

This gives a normalized shock burden

$$B_{\text{shock}} = ap = 0.62 \cdot 0.19865 = 0.123163.$$

Thus, even for a substantially more complicated reentrant configuration, successful termination was achieved while directly stimulating less than 20% of the domain.

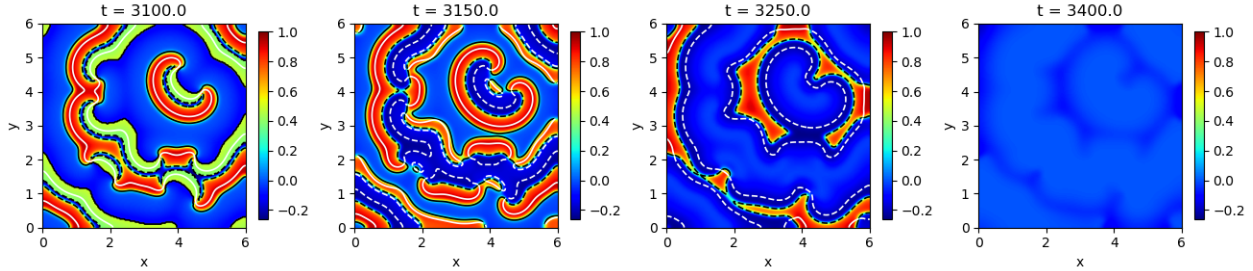


Figure 8: Targeted defibrillation of a complex reentrant wave pattern. The applied stimulus has amplitude 0.62 and covers 19.865% of the domain, giving normalized shock burden $B_{\text{shock}} = 0.123163$.

This result is important because it shows that the targeted method is not merely a convenient way to remove a single clean spiral in an idealized setting. Even when the wave pattern is generated by the more biologically motivated refractory-back mechanism and evolves into a more complicated reentrant configuration, the contour-based intervention remains effective. The success of the method therefore depends less on visual simplicity of the pattern and more on whether the relevant refractory-back geometry is correctly identified and stimulated.

4.2 Varying of Model Parameters

We now will change each of the model parameters and see how they influence the evolution of the biological model. All other parameters are kept the same except for changing the time of the abnormal pulse that initiates spiral waves (based on propagation speed and thickness).

4.2.1 Varying α

The parameter α appears in the cubic ionic current term

$$I_{\text{ion}}(u, v) = u(u - \alpha)(u - 1) + v,$$

so changing α alters the shape of the local reaction kinetics and therefore the excitability of the medium. In qualitative terms, varying α changes how easily an applied perturbation is able to trigger and sustain a propagating excitation. Since spiral formation depends on both successful wave propagation and the subsequent interaction of excitation and recovery, changes in α have a direct effect on whether spiral waves can form at all and, if they do form, whether they can be maintained on the computational domain.

We tested three values of α , namely

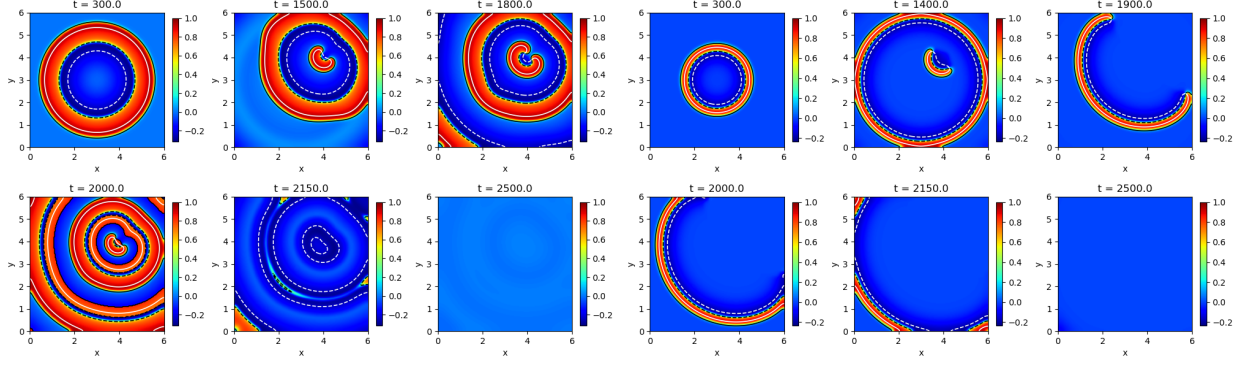
$$\alpha = 0, \quad \alpha = 0.15, \quad \alpha = 0.2,$$

and observed three distinct qualitative regimes.

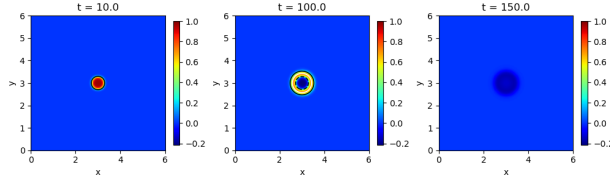
For $\alpha = 0$, spiral waves were readily generated and were noticeably tighter than in the baseline case. In addition, spiral initiation occurred over a much wider range of stimulus timings.

For $\alpha = 0.15$, a spiral wave was still initiated, but its curl was too large to be stably supported within the computational domain. As a result, the spiral gradually drifted outward and eventually disappeared.

For $\alpha = 0.2$, the initial impulses were unable to propagate through the medium at all. Instead of forming traveling waves that could later curl into spirals, the perturbations decayed locally.



(a) $\alpha = 0$, $\beta = 0.3$, $\gamma = 1$, $\delta = 0$, $\epsilon = .01$, $D = .0003$ (b) $\alpha = 0.15$, $\beta = 0.3$, $\gamma = 1$, $\delta = 0$, $\epsilon = .01$, $D = .0003$



(c) $\alpha = 0.2$, $\beta = 0.3$, $\gamma = 1$, $\delta = 0$, $\epsilon = .01$, $D = .0003$

Figure 9: Varying α

4.2.2 Varying β

The parameter β appears in the recovery equation

$$\frac{\partial v}{\partial t} = \epsilon(\beta u - \gamma v - \delta),$$

where it controls how strongly the excitation variable u drives the recovery variable v . In the context of this model, varying β changes how recovery responds to excitation and therefore changes whether the medium settles into persistent excitation, supports traveling waves, or fails to propagate altogether.

We tested three values of β ,

$$\beta = 0, \quad \beta = 0.19, \quad \beta = 0.53,$$

and observed three distinct qualitative behaviors.

For $\beta = 0$, excited cells did not relax back in a way that supported normal wave recovery, and the entire domain approached a uniformly excited equilibrium state.

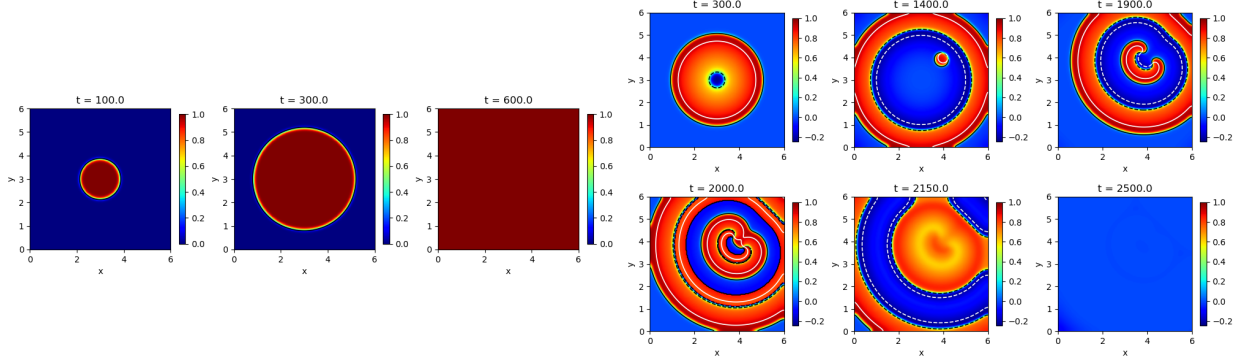
For $\beta = 0.19$, traveling waves and defibrillation behavior reappeared. Propagating waves were sustained, and as β increased within this regime, the wave fronts became noticeably thinner.

For $\beta \geq 0.53$, the initial impulses failed to propagate through the medium. No sustained traveling wave or spiral wave formed.

4.2.3 Varying γ

The parameter γ appears in the recovery equation

$$\frac{\partial v}{\partial t} = \epsilon(\beta u - \gamma v - \delta),$$



(a) $\alpha = 0.1, \beta = 0, \gamma = 1, \delta = 0, \epsilon = .01, D = .0003$ (b) $\alpha = 0.1, \beta = 0.19, \gamma = 1, \delta = 0, \epsilon = .01, D = .0003$

Figure 10: Varying β

where it controls the strength of the negative feedback from the recovery variable v onto itself. In the context of this model, varying γ changes how quickly the recovery variable relaxes and therefore changes how long the medium remains refractory after excitation.

We tested three values of γ ,

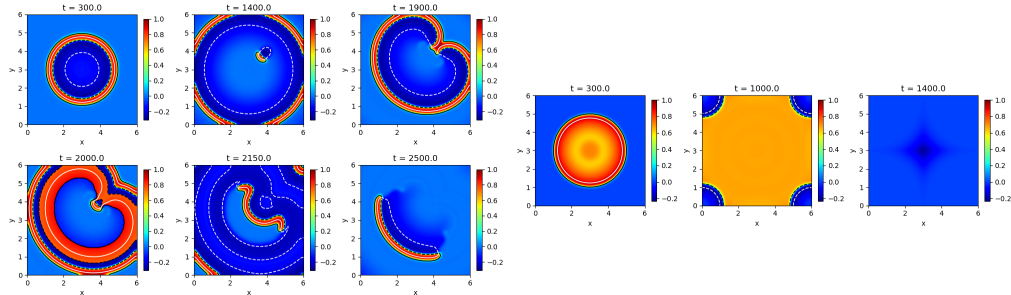
$$\gamma = 0, \quad \gamma = 0.5, \quad \gamma = 1.6,$$

and observed three distinct qualitative behaviors.

For $\gamma = 0$, cells took much longer to recover after excitation. Spiral waves were much easier to form, and targeted defibrillation was not successful.

For $\gamma = 0.5$, recovery became fast enough that defibrillation behavior reappeared and the targeted stimulus was again able to terminate the reentrant activity.

For $\gamma = 1.6$, the waves no longer recovered in a way that supported ordinary propagation. The entire domain became excited, but this excited state was not sustained, and the system eventually collapsed back to a resting equilibrium.



(a) $\alpha = 0.1, \beta = 0.3, \gamma = 0, \delta = 0, \epsilon = .01, D = .0003$ (b) $\alpha = 0.1, \beta = 0.3, \gamma = 1.6, \delta = 0, \epsilon = .01, D = .0003$

Figure 11: Varying γ

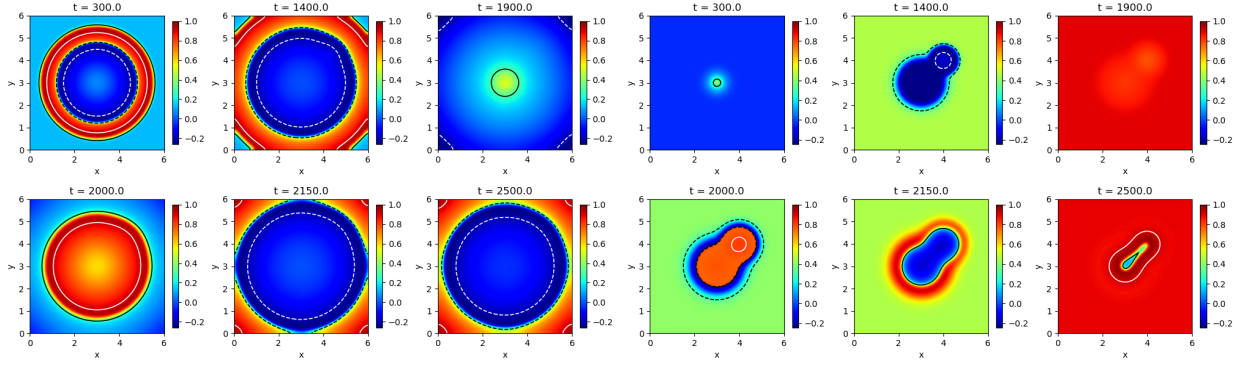
4.2.4 Varying δ

The parameter δ appears in the recovery equation as a constant offset term in the recovery dynamics. In the context of this model, varying δ shifts the baseline balance between excitation and recovery, so changing it can alter whether the medium supports wave propagation, oscillatory behavior, or spatially uniform equilibria.

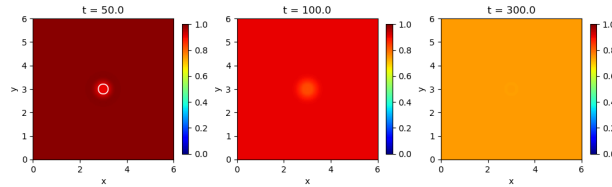
When δ was increased above its baseline value, the system no longer supported the same localized wave behavior seen in the earlier parameter regimes. Instead, the entire domain began to oscillate in a spatially widespread way.

For values of δ above approximately $\delta = 0.08$, these oscillations were replaced by a spatially uniform excited steady state across the whole domain.

Within the FitzHugh-Nagumo model, this means that increasing δ drives the system away from a regime of propagating excitable waves and into regimes dominated first by global oscillation and then by a globally excited equilibrium. The direct biological interpretation of this parameter shift is less clear than for some of the other parameters, but numerically its effect on the wave dynamics is very pronounced.



(a) $\alpha = 0.1, \beta = 0.3, \gamma = 1, \delta = 0.02, \epsilon = .01, D = .0003$ (b) $\alpha = 0.1, \beta = 0.3, \gamma = 1, \delta = 0.5, \epsilon = .01, D = .0003$



(c) $\alpha = 0.1, \beta = 0.3, \gamma = 1, \delta = 0.1, \epsilon = .01, D = .0003$

Figure 12: Varying δ

4.2.5 Varying ϵ

Changing ϵ rescales the recovery dynamics as a whole, preserving the relative contributions of β , γ , and δ while changing the timescale on which the recovery variable evolves. As ϵ is varied, the thickness of the wave and the trajectory traced by the spiral tip both change. For large ϵ , recovery becomes very rapid and disturbances may decay before sustained propagation develops, although for sufficiently large values this behavior may also be influenced by the fixed time step. For very small ϵ , the recovery variable evolves extremely slowly, so the medium may fail to recover on the simulation timescale and the excitation variable can remain near an excited state for long times.

4.2.6 Varying D

Changing the diffusion coefficient D does not meaningfully change the behavior, it simply changes the size of the domain that is of interest because the speed of diffusion is increased.

5 Biological Conclusions and Extensions

The reproduction results show that the main topological ideas of the reference paper are preserved in the FitzHugh–Nagumo setting used here. Spiral waves can be generated either by the biologically motivated refractory-back mechanism or by a symmetry-breaking reset construction, and once formed, their singularities can be moved by stimulating along the refractory back. This teleportation mechanism extends naturally to targeted defibrillation, where only a relatively small fraction of the domain must be stimulated in order to eliminate reentrant activity. The same strategy also remained effective for more complicated multispiral states, indicating that the relevant feature is not the visual simplicity of the wave pattern but the contour geometry supporting the singularities.

The parameter-variation results show more specifically how sensitive this reentrant behavior is to the balance between excitation and recovery. Decreasing α produced tighter spirals and broadened the range of stimulus timings that could initiate reentry, while increasing α first produced spirals too large to remain supported in the domain and then prevented propagation altogether. Changing β moved the system between a uniformly excited state, a regime of propagating thin waves, and a regime in which impulses no longer propagated. Changing γ controlled how quickly the medium recovered after excitation, with slow recovery making spiral formation easier and preventing successful defibrillation, intermediate recovery allowing targeted defibrillation again, and larger values driving the domain through global excitation before collapse to rest. Increasing δ replaced localized wave behavior with whole-domain oscillation and eventually a spatially uniform excited state, while varying ϵ changed the recovery timescale itself, altering wave thickness, spiral-tip motion, and in extreme cases producing either very rapid decay or long-lived excited behavior. Taken together, these results show that whether reentry forms, persists, and can be removed depends strongly on the dynamical regime of the tissue and not only on the geometry of the intervention.

Several natural extensions follow from this work. One is to repeat the same targeted-defibrillation study in more detailed cardiac models, where the variables and parameters have a more direct physiological interpretation. Another is to move beyond two-dimensional domains and test whether the same contour-based ideas extend to three-dimensional scroll-wave dynamics. A third is to systematize the parameter study into a fuller phase-diagram analysis, identifying regions of parameter space associated with stable propagation, spiral formation, spontaneous decay, global oscillation, and successful low-burden control. These directions would help clarify both how robust the topological defibrillation mechanism is and how closely it can be connected to realistic cardiac dynamics.

6 Contributions

Zane Perry primarily developed, implemented, and tested the simulation code used in this project, including the numerical workflow for wave initiation, spiral evolution, and defibrillation experiments, as well as the corresponding figures. Anil Ambrosi primarily developed the mathematical analysis and interpretation used in the report and carried out much of the testing across different parameter regimes, and corresponding figures. Both authors worked together on the preparation of the final presentation.

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